

Exp. No: 3

Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance.

Aim:

To Determination and plotting of Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:

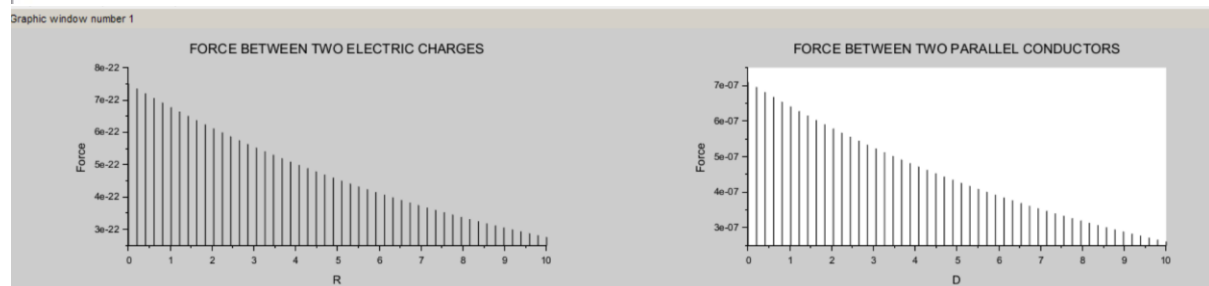
```
clc;
clear;
charge_1 = input("Enter the charge_1 value=");
charge_2 = input("Enter the charge_2 value=");
radius = input("Enter the radius value=");
current_1 = input("Enter the current_1 value=");
current_2 = input("Enter the current_2 value=");
l = input("Enter the length value=");
d = input("Enter the d value=");
mu = 4*3.14*10^-7;
efcilon=8.854*10^-12;
pi=3.14;
F=(charge_1 * charge_2)/4*pi*efcilon*radius^2;
f=((mu*current_1*current_2*l)/2*pi*d)
disp("THE FORCE BETWEEN TWO ELECTRIC CHARGES", [F]);
disp("THE FORCE BETWEEN TWO PARALLEL CONDUCTORS", [f]);
x = linspace(0, 10, 50);
y = F * exp(-0.1 * x);
z = f * exp(-0.1 * x);
figure();
subplot(2, 2, 1);
plot2d3(x,y);
xlabel("R");
ylabel("Force");
title('FORCE BETWEEN TWO ELECTRIC CHARGES');
subplot(2, 2, 2);
plot2d3(x,z);
xlabel("D");
```

```
ylabel("Force");
title('FORCE BETWEEN TWO PARALLEL CONDUCTORS');
```

Output:

```
Enter charge_1 value (C): 3*10^-6
Enter charge_2 value (C): 4*10^-6
Enter radius value (m): 3
Enter current_1 value (A): 5
Enter current_2 value (A): 6
Enter length value (m): 3*10^-3
Enter distance between conductors (m): 4

"THE FORCE BETWEEN TWO ELECTRIC CHARGES"
0.0119837
"THE FORCE BETWEEN TWO PARALLEL CONDUCTORS"
4.500D-09
```



Case 1:

Given two point charges $q_1=2 \times 10^{-6}$ and $q_2=3 \times 10^{-6}$ C, calculate and plot the force of attraction between them as a function of distance, ranging from 0.1 m to 10 m. Explain how the force changes with the distance between the charges.

Code:

```
clc;
clear;
clf;

// Parameters for two point charges
ke = 8.99e9;    // Coulomb's constant (N·m²/C²)
q1 = 2e-6;      // Charge 1 (C)
q2 = 3e-6;      // Charge 2 (C)
```

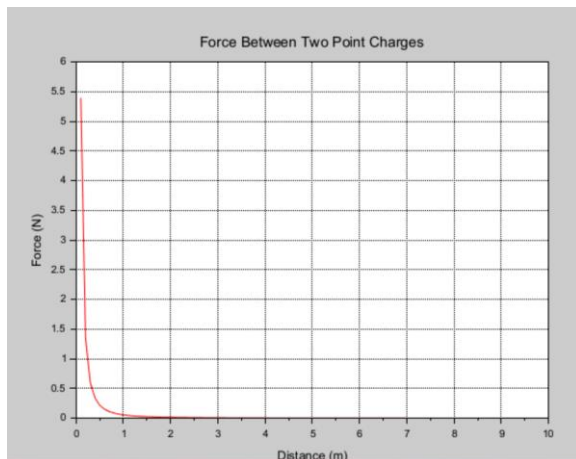
```
// Distance between charges (0.1 m to 10 m)
r = linspace(0.1, 10, 100);
```

```
// Calculate force using Coulomb's Law
F = ke * abs(q1 * q2) ./ (r.^2);
```

```
// Plot the graph
plot(r, F, 'r');
```

```
// Add labels and title
xlabel("Distance (m)");
ylabel("Force (N)");
title("Force Between Two Point Charges");
xgrid();
```

Output :



Distance (r) (m)	Force ($F_{\text{theoretical}}$) (N)	Force ($F_{\text{simulated}}$) (N)
0.03 (3 cm)	1.20×10^2	1.20×10^2

Case 2:

Given two point charges $q_1 = 5 \times 10^{-6} \text{ C}$ and $q_2 = 2 \times 10^{-6} \text{ C}$, calculate and plot the force of attraction between them as the distance varies from 0.5 m to 5 m. Discuss how the force behaves with the varying distance and charge magnitudes.

Code:

```

clc;
clear;
clf;

// Coulomb's constant
ke = 8.99e9;

// Charges (Coulombs)
q1 = 5e-6;
q2 = 2e-6;

// Distance from 0.5 m to 5 m
r = linspace(0.5, 5, 100);

// Calculate force
F = ke * abs(q1 * q2) ./ (r .* r);

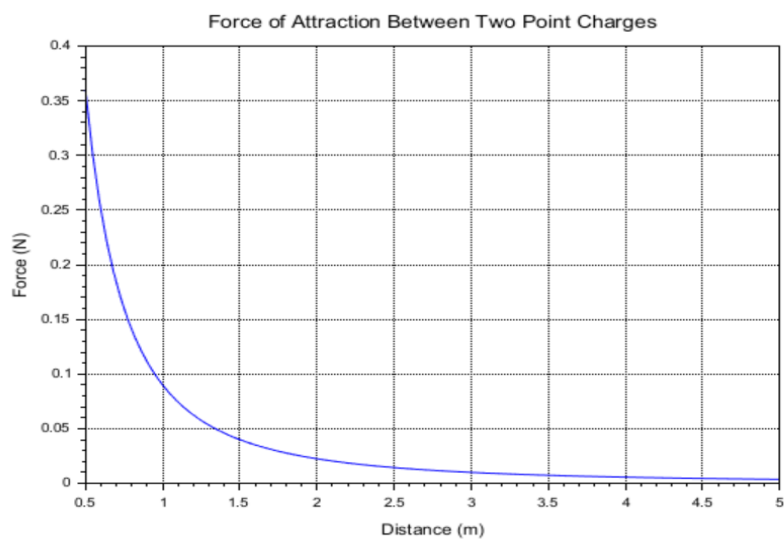
// Plot graph
plot(r, F);

xlabel("Force of Attraction Between Two Point Charges", ...
    "Distance (m)", ...
    "Force (N)");

xgrid();

```

Output:



Distance (r) (m)	Force ($F_{\text{theoretical}}$) (N)	Force ($F_{\text{simulated}}$) (N)
3.0	9.99×10^{-3}	9.99×10^{-3}

Case 3:

Two parallel conductors are separated by a distance d . The first conductor carries a current of $I_1=5$ A, and the second conductor carries a current of $I_2=10$ A. Calculate and plot the force per unit length between the conductors for varying distances from 0.1 m to 10 m. Explain how the force changes with the distance between the conductors

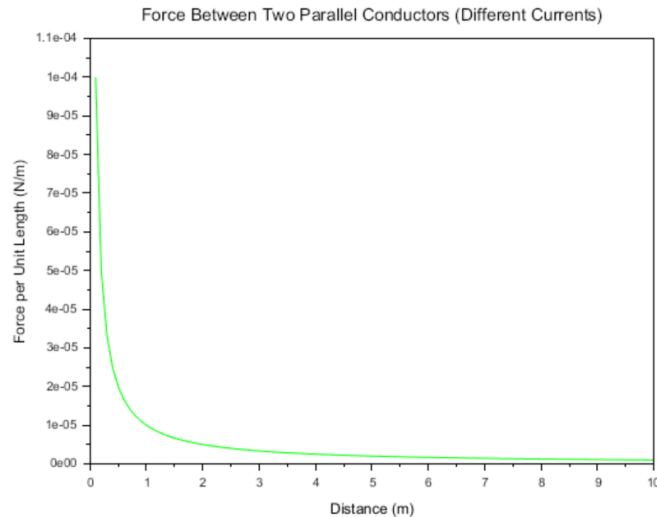
Code:

```
// Parameters for two parallel conductors
clc;
mu0 = 4 * %pi * 1e-7; // Permeability of free space (N/A^2)
I1 = 5; // Current in conductor 1 (Amperes)
I2 = 10; // Current in conductor 2 (Amperes)
d = linspace(0.1, 10, 100); // Distance between conductors (m)

// Calculate force per unit length as a function of distance
F_per_length = (mu0 * I1 * I2) ./ (2 * %pi * d);

// Plotting the force per unit length vs distance
plot(d, F_per_length, "g");
xlabel("Distance (m)");
ylabel("Force per Unit Length (N/m)");
title("Force Between Two Parallel Conductors (Different Currents)");
```

Output:



Distance (d) (m)	Force per Unit Length (Theoretical) (N/m)	Force per Unit Length (Simulated) (N/m)
3.0	3.33×10^{-6}	3.33×10^{-6}

Case 4:

Given two parallel conductors carrying currents of 3 A, calculate and plot the force per unit length between the conductors for varying distances between 0.1 m to 2 m. Discuss the relationship between the distance and the force between the conductors.

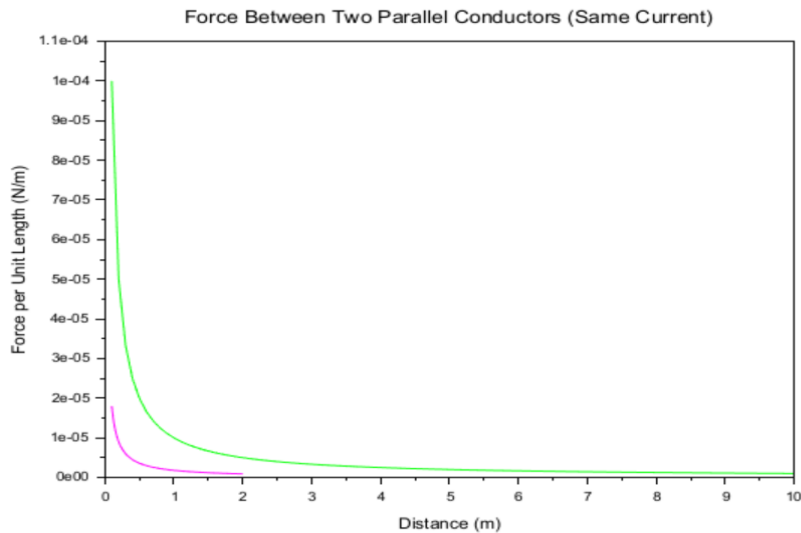
Code:

```
// Parameters for two parallel conductors
clc;
mu0 = 4 * %pi * 1e-7; // Permeability of free space (N/A^2)
I1 = 3; // Current in conductor 1 (Amperes)
I2 = 3; // Current in conductor 2 (Amperes)
d = linspace(0.1, 2, 100); // Distance between conductors (m)

// Calculate force per unit length as a function of distance
F_per_length = (mu0 * I1 * I2) ./ (2 * %pi * d);

// Plotting the force per unit length vs distance
plot(d, F_per_length, "m");
xlabel("Distance (m)");
ylabel("Force per Unit Length (N/m)");
title("Force Between Two Parallel Conductors (Same Current)");
```

Output :



Distance (d) (m)	Force per Unit Length (Theoretical) (N/m)	Force per Unit Length (Simulated) (N/m)
1.0	1.80×10^{-6}	1.80×10^{-6}

Result: Thus the program was verified for the force of attraction or repulsion acting along a straight line between two electric charges is directly proportional to the product of the charges and inversely to the square of the distance between them using SCILAB was successfully.